

VU Student Attendance Reporting

Complete Instruction Manual (End User + Admin Guide)

> Version: Repository state as of this document commit

> Applies to: `master` and `release` branches

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Purpose of this application

VU Student Attendance Reporting is a Windows desktop application used by teaching and support staff to:

- manage attendance-related interventions
- send structured communication emails to students/employers
- monitor student unit progression
- handle investigation/amendment workflows
- track Exemplar profiling progress
- keep reporting and audit data synchronized with SQL-backed records

The design goal is to centralize student operations in one desktop workflow instead of splitting work across separate systems.

What the application can do (feature summary)

Core capabilities

1. ****Student selection and context loading****

- filter by block group/class
 - search by Student ID
 - load key student/employer details from SQL
2. ****Attendance/intervention communications****
 - generate Outlook draft emails using SQL-managed templates
 - log relevant attendance counters and report dates in SQL
 - support standard warning/progress/report scenarios
 3. ****Student investigation process****
 - structured prompts for follow-up actions
 - saves outcomes back to database
 - generates investigation communication
 4. ****Unit progression and competency management****
 - view/update per-unit status
 - refresh Exemplar unit progression percentages
 - generate and email authority-to-sit documentation
 5. ****Exemplar profiling integration****
 - API-based student lookup and profiling card summaries
 - optional student-specific profiling email override
 - support for production/staging API modes
 6. ****Administrative operations****
 - manage email settings, templates, teacher data
 - CSV/Excel-driven SQL maintenance operations
 - SQL connection and database date administration
 7. ****Application updating****
 - checks GitHub Releases for newer versions
 - can enforce mandatory update policies
 - launches installer assets directly from release

System requirements and prerequisites

Required

- Windows machine (app is WinForms/.NET desktop)
- SQL connectivity to configured environment
- Microsoft Outlook installed (required for email workflows)
- Microsoft Excel installed (for Excel-integrated operations)

Included by current GitHub release installer

The current GitHub release pipeline packages:

- self-contained app runtime payload
- application files and dependencies
- Exemplar login JAR assets
- bundled Java runtime (`jre\bin\java.exe`, Temurin 17) for Exemplar login compatibility

Not bundled

- Outlook/Excel desktop installations
- organization-specific VPN/network access requirements

Install and first launch

1. Download the latest release installer from GitHub Releases.
2. Run `StudentAttendanceReporting-Setup.msi`.
3. Launch using Start Menu or Desktop shortcut (created by installer).
4. On first run:
 - verify SQL connection is available
 - verify Outlook is installed and opens
 - verify student data loads via class selection or student ID search

If SQL is unavailable, the app will display a SQL error workflow and allow connection updates (admin paths).

Main screen user guide

Typical daily workflow

1. Open app.
2. Select **BlockGroup/Class Name** (or use **Search Student ID**).
3. Select student from list.
4. Review loaded context (student/employer, attendance/profiling labels).
5. Choose required **Email Subject** template.
6. Complete visible fields.
7. Click **Submit** (or **Student Investigation** for that specific workflow).
8. Optionally open **Check Students Units** for unit/profiling actions.

Main controls (practical meaning)

- **Check for Updates**: triggers update check and optional installer launch.
- **Search Student ID + Search**: direct student lookup path.
- **BlockGroup/Class Name** dropdown: class-driven student selection path.
- **Email Subject**: selects communication scenario and adjusts form fields.
- **Submit**: validates and builds Outlook draft based on selected template.
- **Student Investigation**: dedicated investigation flow (only for that template).

- ****Check Students Units****: opens Student Units screen.
- ****BlockGroup Email****: class-wide email utility.
- ****Amend/Assign Exemplar profiling email****: saves/removes student-specific Exemplar email override.
- ****Student Re-Allocation Request****: opens re-allocation request flow (link + email draft).
- ****Issue Report / Feature Request****: opens external reporting form.

Email workflows and template behavior

Email bodies/help content are managed from SQL templates. Users select template via ****Email Subject****.

Common template options

- Student Term Progress Report
- 2 Week Intention Letter
- 4 Week Intention Letter
- Course Withdraw Notice
- Student Behaviour Notice
- Overdue Fees - Warning
- Overdue Fees - Sanction
- Unit Withdraw Notice
- Absent Notice
- Late Arrival Notice
- Early Departure Notice
- Sent Back to Work Notice
- Student Unit Report
- Student Investigation
- Class Commencement Reminder
- Yearly Student Report
- Exemplar Profiling Outstanding Alert

> Notes:

- > - Exact list displayed comes from SQL template table.
- > - Some fields become visible/required only for specific subjects.

Submit behavior

When ****Submit**** is used:

- app validates visible/required fields
- builds template body from SQL with variable replacement
- appends signature and contextual status content as needed
- opens an Outlook draft with recipients/subject/body

Investigation behavior

For **Student Investigation**:

- use **Student Investigation** button (not standard Submit path)
- prompts user through defined contact/actions flow
- logs outcomes to SQL
- prepares corresponding email communication

Student Units screen guide

Open via **Check Students Units**.

What it is for

- monitor and manage student unit completion/progression
- check profiling percentages
- produce and send authority-to-sit documentation

Key actions

1. **Refresh Profiling %**
 - pulls latest unit progression from Exemplar endpoints
 - updates unit-level experience/demonstration displays
2. **Unit checkboxes**
 - update SQL-backed completion state
 - protected by admin override prompt for direct modifications
3. **Generate Authority to sit letter**
 - produces LEA/assessment letter output
4. **Email Authority to sit letter**
 - opens prepared Outlook draft with generated letter attachment
5. **Email Profiling Correspondence**
 - creates email draft for insufficient card/profiling communication

Student Amendment workflow

The Student Amendment form supports class assignment changes and related communications.

Supported actions

- add student to blockgroup
- move student between blockgroups
- remove student from current blockgroup
- report incorrect employer details

Inputs typically required

- current/proposed blockgroup
- current/proposed class teacher
- reason for change
- proposed effective/start date
- submitter identity

Output

- structured Outlook draft(s) to relevant stakeholders

Settings and admin tools

Admin access is controlled through the admin entry path on main screen.

Settings capabilities

- manage notification email addresses (Admin/App-Train/Trades)
- upload or update signature image
- manage teacher list
- open Email Templates management
- run data import/update operations (admin variants)

Email Templates management

- select existing template
- edit subject/body/help text
- add/delete/reset template entries

Advanced/admin maintenance (SettingsForm/Admin)

- update SQL connection string and restart
- update database date metadata
- run student/unit maintenance tasks
- upload agreements and perform SQL housekeeping actions

Exemplar profiling integration guide

What is integrated

- student lookup in Exemplar
- card summary/status retrieval
- unit progression endpoint calls
- optional qualification status update support in code

Runtime behavior

- uses configured API base URL and token flow
- login JAR is selected by environment (production/staging logic)
- app can store student-specific override email in SQL

Important for users

- if profiling is unavailable, app reports reason in status area
- with bundled Java, most Java runtime issues should be removed on release builds

Updates and release behavior

The app checks GitHub Releases (`releases/latest`), not branches.

Update becomes visible to users only when:

1. a newer release tag is published (e.g., `v3.1`, `v3.2`)
2. release has installer assets attached (`.msi`/`.exe`)

Policy controls (release notes metadata)

Optional keys can enforce update behavior:

- `min\_required\_tag`
- `force\_update`
- `asset\_name`

Troubleshooting guide

1) SQL Connection Error

Symptoms:

- startup SQL error screen
- data not loading

Actions:

1. verify network/VPN access
2. verify SQL server reachability
3. update SQL connection string (admin path)
4. restart app

2) Outlook/email issues

Symptoms:

- email draft cannot open
- Outlook automation errors

Actions:

1. verify Outlook is installed and opens manually
2. ensure default profile is configured
3. retry Submit after Outlook is ready

3) Exemplar profiling shows “Not configured” or login failure

Actions:

1. verify Exemplar settings and environment pairing (prod vs staging)
2. verify correct login JAR selection
3. verify bundled Java exists (`jre\bin\java.exe`) in install folder
4. verify credentials/token configuration path

4) Update check not offering update

Actions:

1. verify release is actually published (not draft)
2. verify release tag is newer than installed major/minor
3. verify assets are attached

5) Unit/profiling values do not refresh

Actions:

1. click ****Refresh Profiling %****
2. verify student has valid mapping data for requested unit set
3. verify Exemplar API connectivity and qualification configuration

Operational notes for IT/admins

Branch and release model

- `master`: development baseline
- `release`: branch that triggers automated build/release workflow

Current release automation includes

- version tag management (`vX.Y`)
- self-contained app publish
- WiX MSI build
- shortcut/icon setup
- Java runtime bundling
- GitHub Release publish with installer assets

Recommended governance

- maintain template quality in SQL `EmailTemplates`
- regularly validate email routing addresses in settings
- validate SQL schema/table availability after environment changes
- keep Exemplar production/staging config aligned with intended JAR/token flow

Quick reference: where to find more detail

- Release automation: `docs/RELEASE-AUTOMATION.md`
- MSI packaging details: `docs/MSI-Installer-Guide.md`
- Exemplar setup and debugging: `EXEMPLAR\_API\_SETUP.md`
- OTA policy options: `OTA\_RELEASE\_POLICY.md`
- Branch process notes: `WORKFLOW.md`

End of manual